Does Baseline Spiking Activity of a Neuron Enhance its Data Transmission Rates?

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January 8, 2024

Abstract

In this note, the authors propose an open problem in computational neuroscience.

It is well known that there is a dark current, or noisy substratum, in the retinal neurons. That is, these neurons fire even in the absence of a visual stimulus. What might be the purpose of this firing activity? We hope to work towards an answer to this question. In this context, consider the following quotation [1]:

> Researchers into quantum information and open quantum systems are realizing that in some situations, noise can in fact be used to aid in information processing tasks. In this work, we have introduced a technique whereby a generalized type of Markovian quantum process can be used to aid in the preservation of quantum information.

To emulate the development in the source paper, we may start with a Markovian channel: take the Poisson channel, for example and identify the ideal codebook. Then we can quantify the worst-case performance of the channel over all codewords by defining a fidelity measure. Next, on top of this Poisson noise, we may add a "generalized" Poisson process, or Poisson process with non-trivial memory. For this augmented channel, we repeat the quantification step and check - are there choices of the memory process for which the worst-case performance of the combined system is in fact better than that of the original background system? What about the worst-case performance over just some fixed duration of channel-use time? The answer to these last questions might yield clues to the original question posed in this note.

References

[1] Jeffrey Marshall, Lorenzo Campos Venuti, and Paolo Zanardi. Noise suppression via generalized-markovian processes. *Physical Review A*, 96(5):052113, 2017.